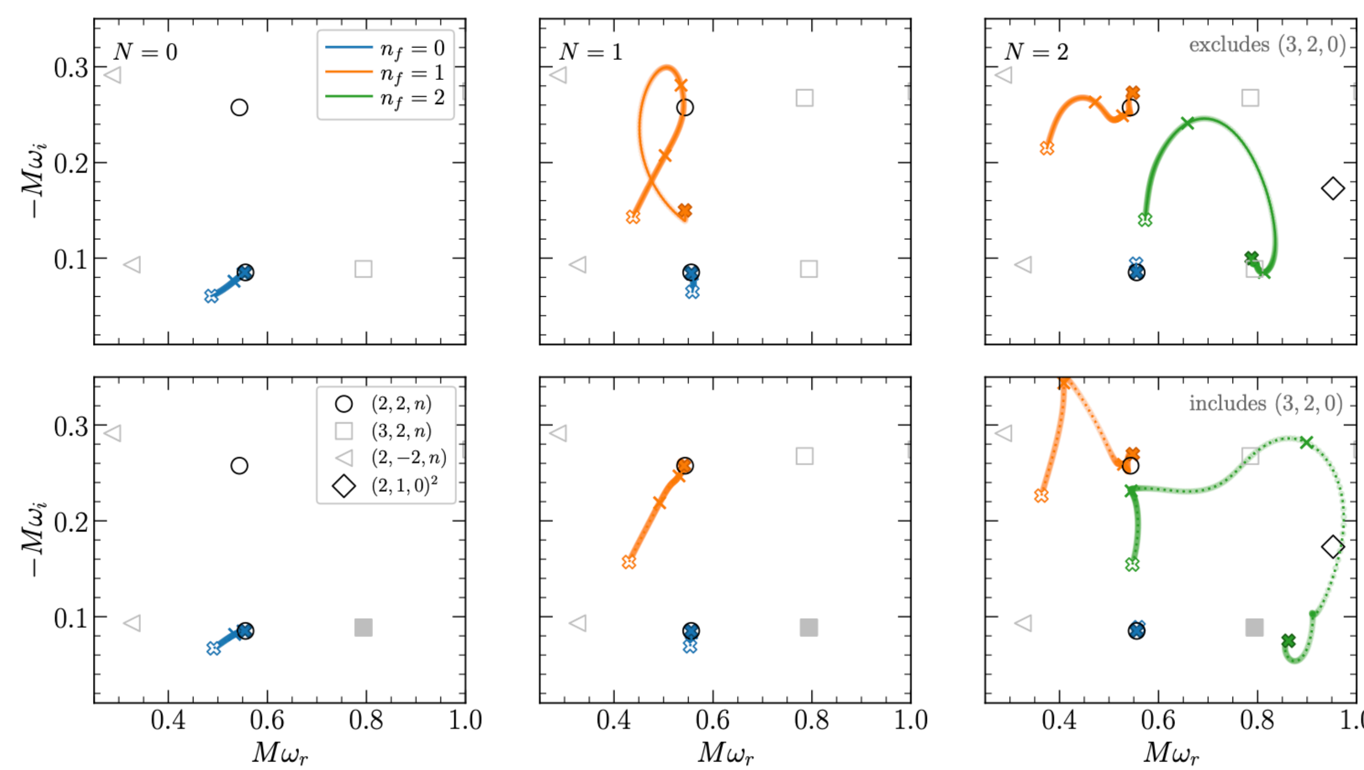
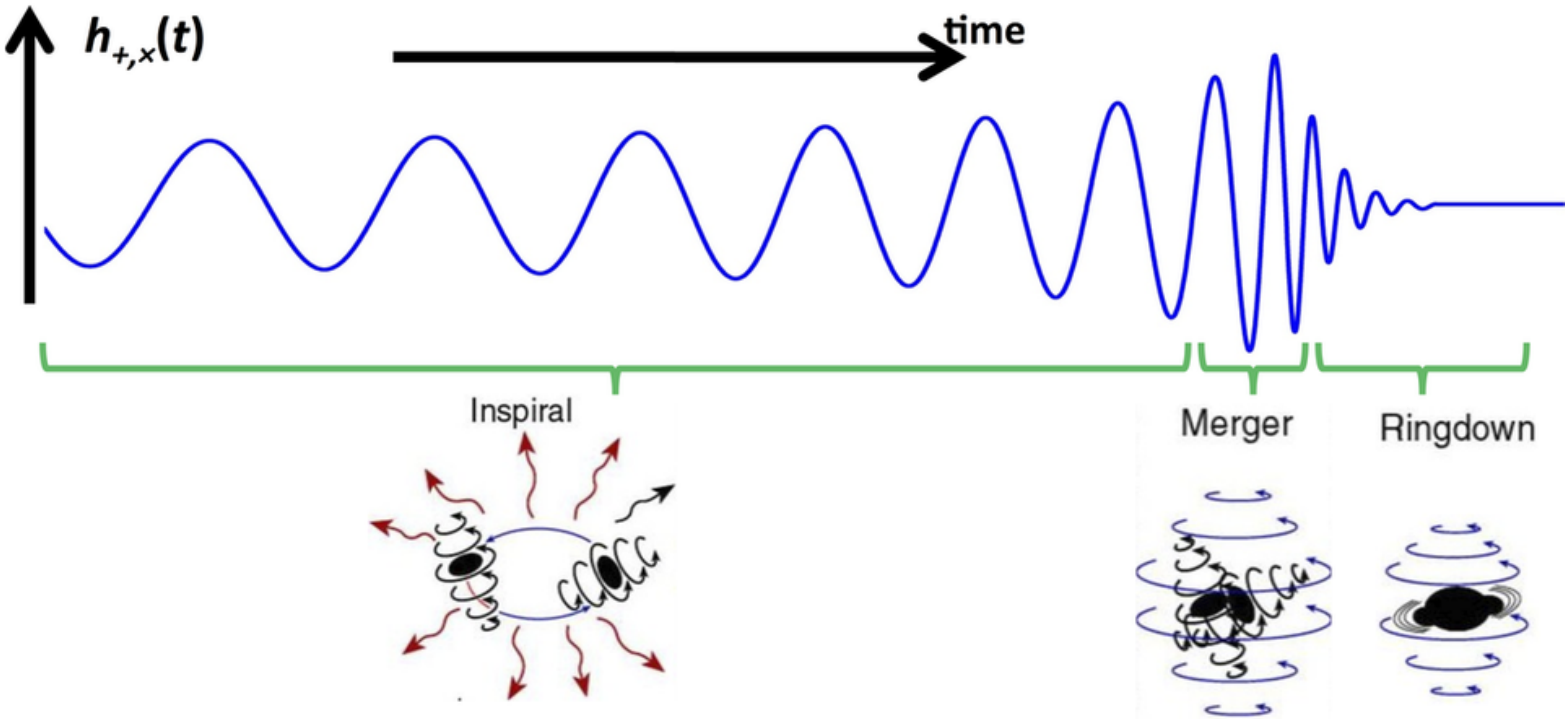
Agnostic black hole spectroscopy: quasinormal mode content of numerical relativity waveforms and limits of validity of linear perturbation theory

Baibhav, Cheung, Berti, Cardoso, Carulla, Cotesta, Del Pozzo, and Duque (2023)
arxiv: 2302.03050

Juan Rodrigo
Gravity Journal Club talk, 4 December 2023



A physical scenario: Coalescing binaries



Gravitational wave detectors are designed to "hear" this process (among other things). We will be concerned with the ringdown phase.

[Figure from soundsofspacetime.org/coalescing-binaries.html]

BH spectroscopy applied to mergers

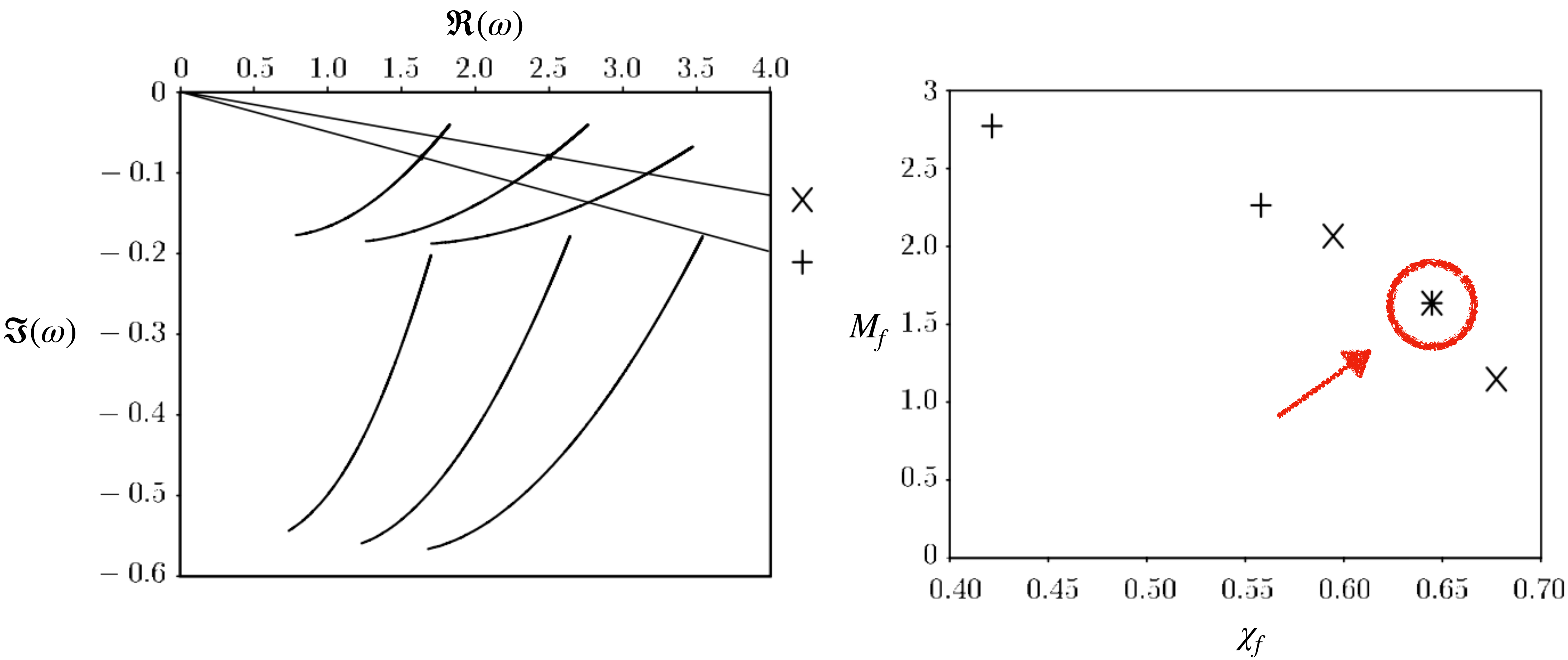
- We expect the gravitational waves emitted in the post-merger ("ringdown") phase of a BBH merger to be dominated, for a certain time window, by a combination of quasinormal modes.

$$h_+ - ih_\times \equiv \sum_{\ell m} h_{\ell m}(t)\,{}_{-2}Y_{\ell m}(\iota, \phi), \qquad h_{\ell m}(t) \equiv \sum_n A_{\ell mn}\, e^{-i[\omega_{\ell mn}(t - t_{start}) + \phi_{\ell mn}]}\,.$$

- The QNM amplitudes and phases depend on the nature of the perturbation, but the complex frequencies are characteristic of the remnant: according to GR, we have $\omega_{\ell mn} = \omega_{\ell mn}(M_f, \chi_f)$. Alternative theories of gravity may predict otherwise.

BH spectroscopy applied to mergers

In principle, measurement of one mode allows extraction of M_f, χ_f, and measurement of two modes allows for a test of the no-hair theorem.



[adapted from Dreyer et al. (2004)]

Can we detect QNM overtones?

- The overtone index n orders the QNM modes for a given (ℓ, m) by damping timescale – the fundamental mode is the longest-lived and corresponds to $n = 0$.

$$h_{\ell m}(t) \equiv \sum_n A_{\ell mn}\, e^{-i[\omega_{\ell mn}(t-t_{start})+\phi_{\ell mn}]}$$

- Generally, overtones are assumed to be subdominant to contributions from other (ℓ, m) modes, since these are short-lived and difficult to identify with confidence in the data.

- Identifying overtones in real data is a controversial topic!

Giesler et al. (2020): including overtones will help!

Black hole ringdown: the importance of overtones

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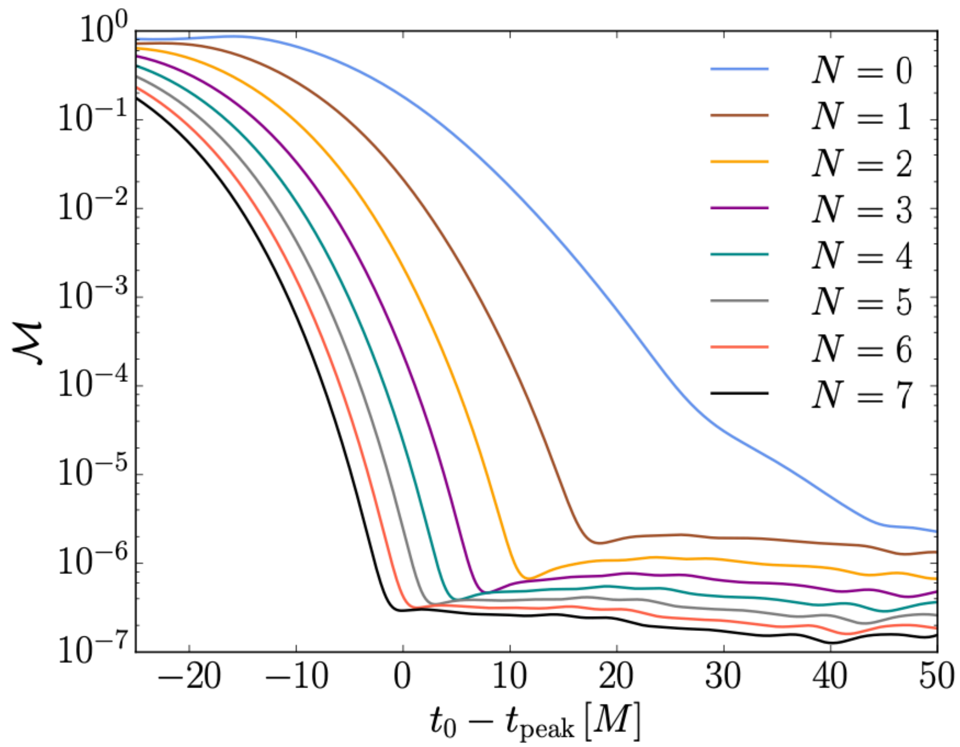
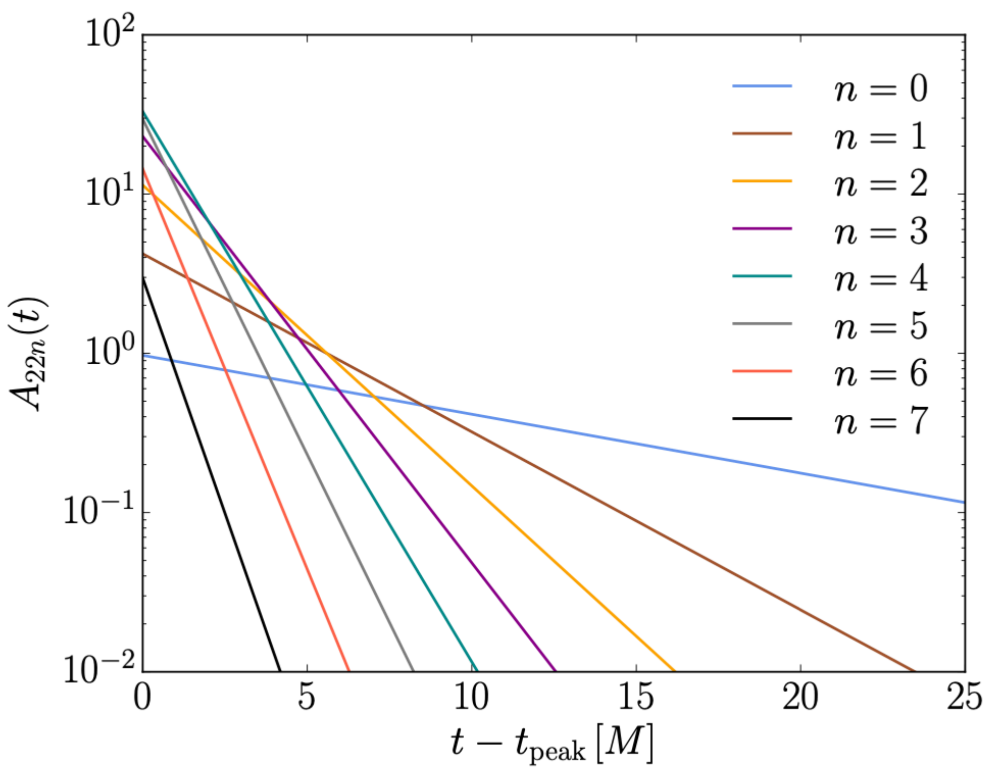
(Dated: January 13, 2020)

It is possible to infer the mass and spin of the remnant black hole from binary black hole mergers by comparing the ringdown gravitational wave signal to results from studies of perturbed Kerr spacetimes. Typically these studies are based on the fundamental quasinormal mode of the dominant $\ell = m = 2$ harmonic. By modeling the ringdown of accurate numerical relativity simulations, we find, in agreement with previous findings, that the fundamental mode alone is insufficient to recover the true underlying mass and spin, unless the analysis is started very late in the ringdown. Including higher overtones associated with this $\ell = m = 2$ harmonic resolves this issue, and provides an unbiased estimate of the true remnant parameters. Further, including overtones allows for the modeling of the ringdown signal for all times beyond the peak strain amplitude, indicating that the linear quasinormal regime starts much sooner than previously expected. This implies that the spacetime is well described as a linearly perturbed black hole with a fixed mass and spin as early as the peak. A model for the ringdown beginning at the peak strain amplitude can exploit the higher signal-to-noise ratio in detectors, reducing uncertainties in the extracted remnant quantities. These results should be taken into consideration when testing the no-hair theorem.

Their claim: perturbation theory is relevant as early as the time of peak strain amplitude, *provided you include overtones.* The present paper is, in part, a response to this idea.

Giesler et al. (2020): including overtones will help!





- The authors worked with the GW150914-like NR waveform SXS:BBH:0305, and found that for the $\ell = m = 2$ mode, increasing the number of overtones used for fitting (up to $N = 7$) caused mismatch with the waveform to decrease.
- They also found that using more overtones allowed for more accurate extraction of the remnant parameters, even when the fitting times started at the peak of strain.
- Finally, they challenge the notion that higher overtones are subdominant to the fundamental mode by computing the corresponding amplitudes at the peak of strain.

An analysis of Giesler et al. (2020)'s claims

Can we reliably conclude from an analysis of NR simulations that the entire post-peak waveform is described by a superposition of overtones consistent with linear BH perturbation theory of a fixed Kerr background?

(Baibhav and collaborators say ***no***.)

Outline

1. Extracting overtones in linearized theory
2. Post-peak BBH waveforms
3. Extracting complex frequencies from NR waveforms

Notation and fitting models

- The waveform is fit `(NonlinearModelFit)` by $N+1$ damped sinusoids

$$Q_N(t) = \sum_{n=0}^{N} A_{\ell mn}\, e^{-i[\omega_{\ell mn}(t-t_{peak})+\phi_{\ell mn}]},$$

  where (assuming GR is the correct description) $\omega_{\ell mn} = \omega_{\ell mn}(M_f, \chi_f)$.

- One can choose to be agnostic about the theory, and allow some or all of these modes to be "free" – i.e. the fitting model makes no predictions about the complex frequencies ω .

Extracting overtones in linearized theory

- As a simpler case, the authors start by considering a linearly perturbed Schwarzschild geometry.

- The wavefunction describing the perturbation is $\mathbf{\Psi}_{num}(r, t)$, obtained by numerically solving the Regge-Wheeler equation with initial data

$$\mathbf{\Psi}_{num}(0, r) = 0, \qquad \frac{\partial \mathbf{\Psi}_{num}}{\partial t}(0, r) = \exp\left[-\frac{(r_* - r_0)^2}{2\sigma^2}\right],$$

then the time-domain waveform $\mathbf{\Psi}_{num}(t)$ is extracted at future null infinity.

Extracting overtones in linearized theory

The result is qualitatively similar to the one pictured here. We identify three distinct parts in the waveform:

1. presence of "prompt response" component depending on initial data,
2. "ringdown" (superposition of QNMs) at intermediate times,
3. a late-time power-law tail.

[Kokkotas and Schmidt (1999), in Living Rev. Relativ.]

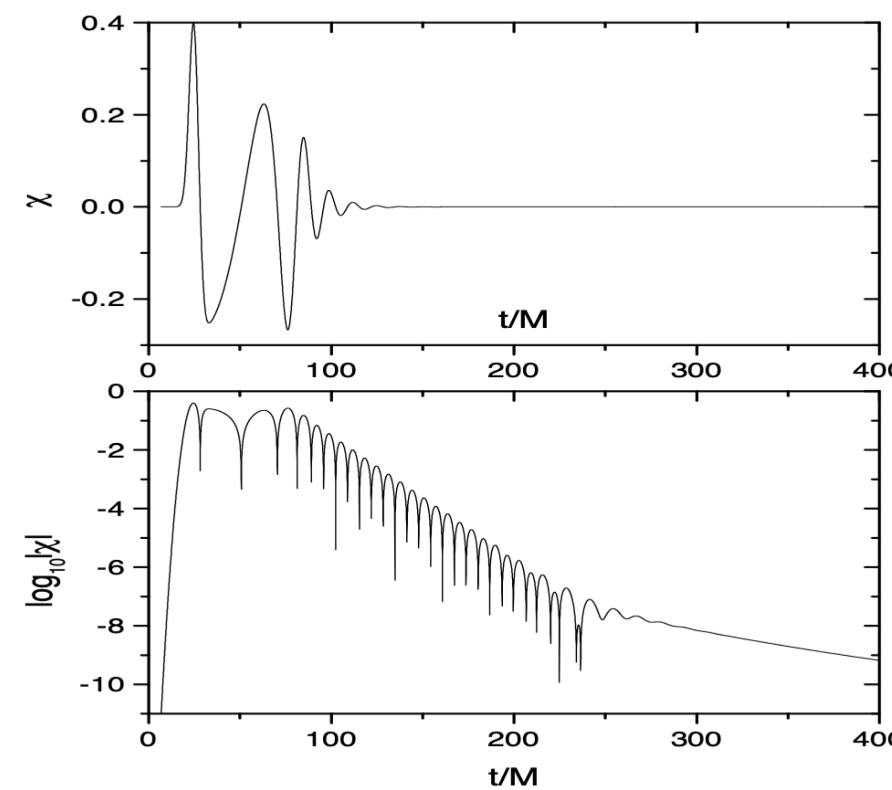


Figure 4: *The response of a Schwarzschild black hole as a Gaussian wave packet impinges upon it. The QNM signal dominates the signal after $t \approx 70M$ while at later times (after $t \approx 300M$) the signal is dominated by a power-law fall-off with time.*

Extracting overtones in linearized theory

- The $\ell = 2$ multipole of the radiation is dominant, and for a nonrotating background all components with $m \neq 0$ vanish:

$$A_{\ell mn} \to A_n \equiv A_{20n} \,.$$

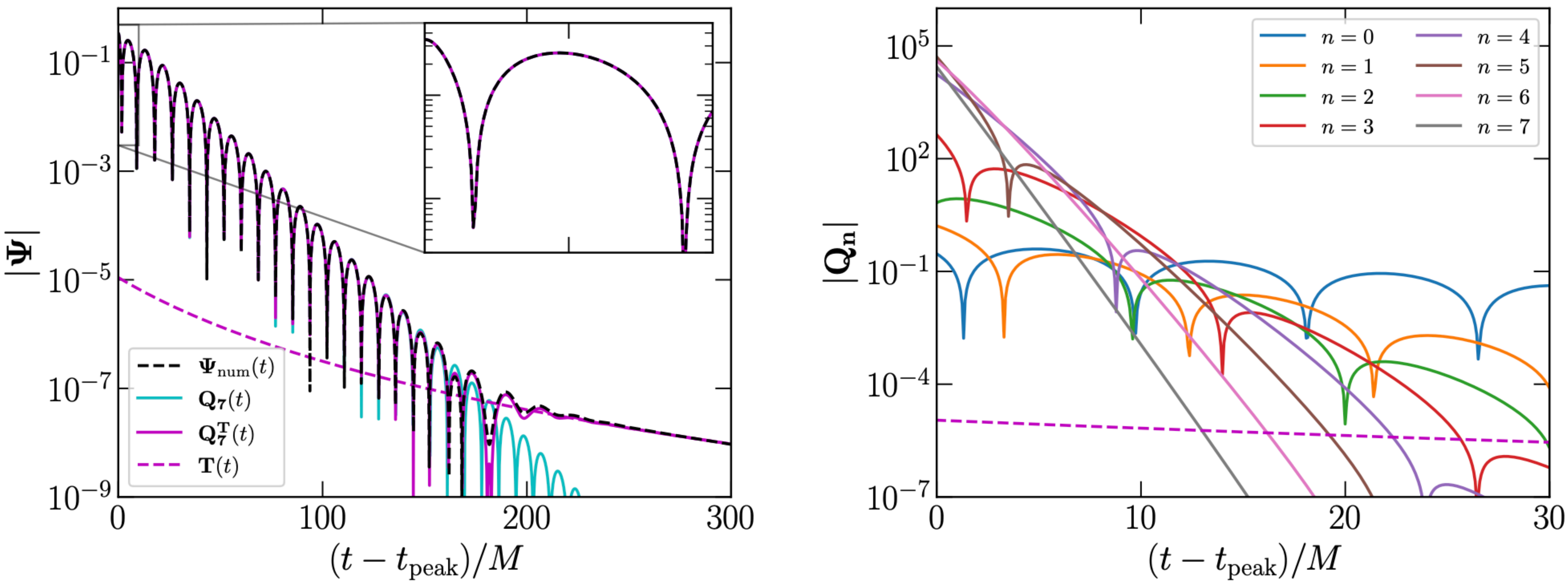
- Applying a similar labeling scheme for ϕ_n and ω_n, the fitting model now reads

$$Q_N(t) \equiv \sum_{n=0}^{N} A_n \, \exp^{-i[\omega_n(t-t_{peak})+\phi_n]}, \qquad\qquad t \in (t_0, t_{end}) \,.$$

  where t_{end} is an arbitrarily chosen (but large) end time.

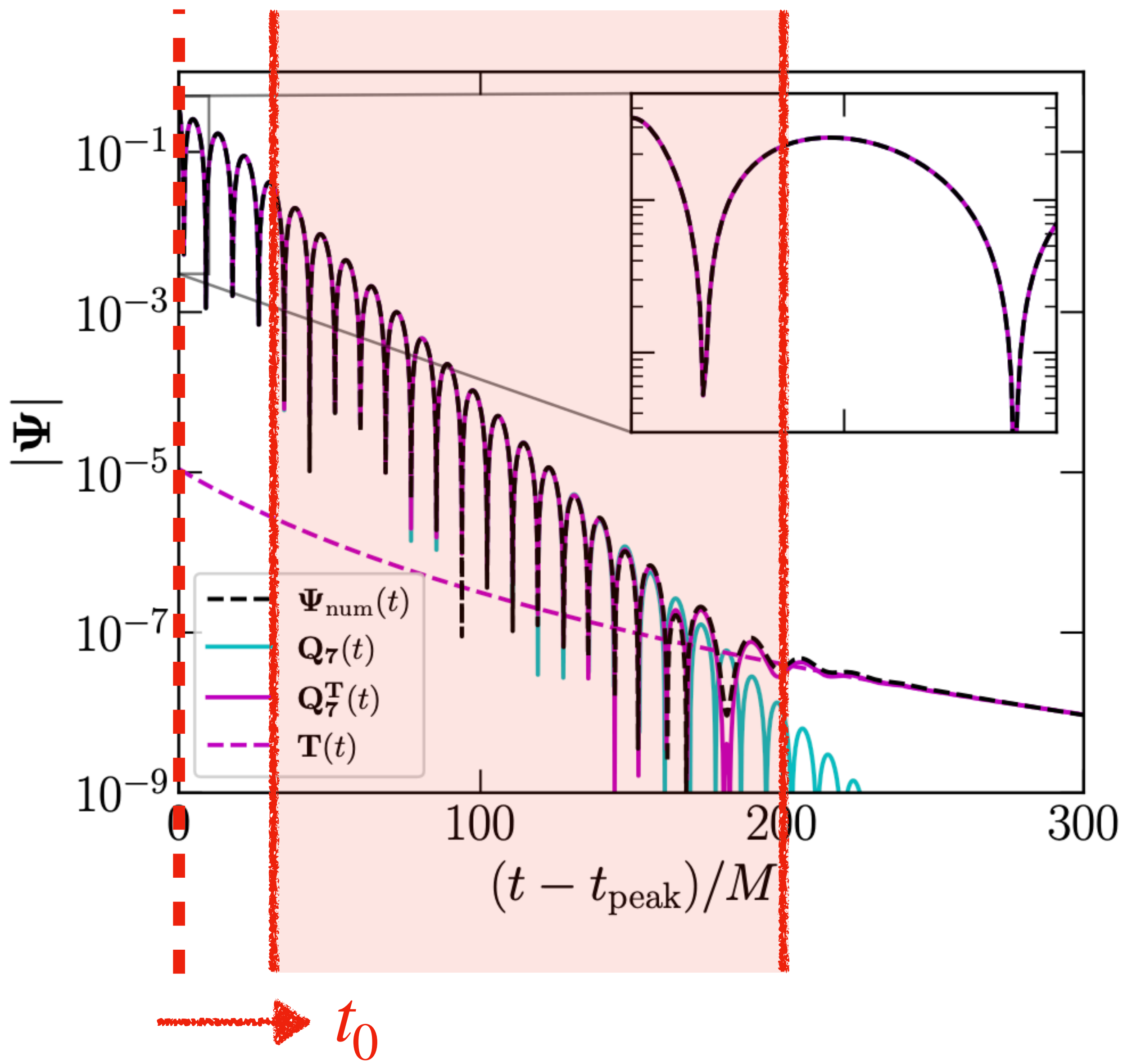
- Waveforms to be fit are denoted in boldface type, e.g. $\mathbf{\Psi}_{num}(t)$.

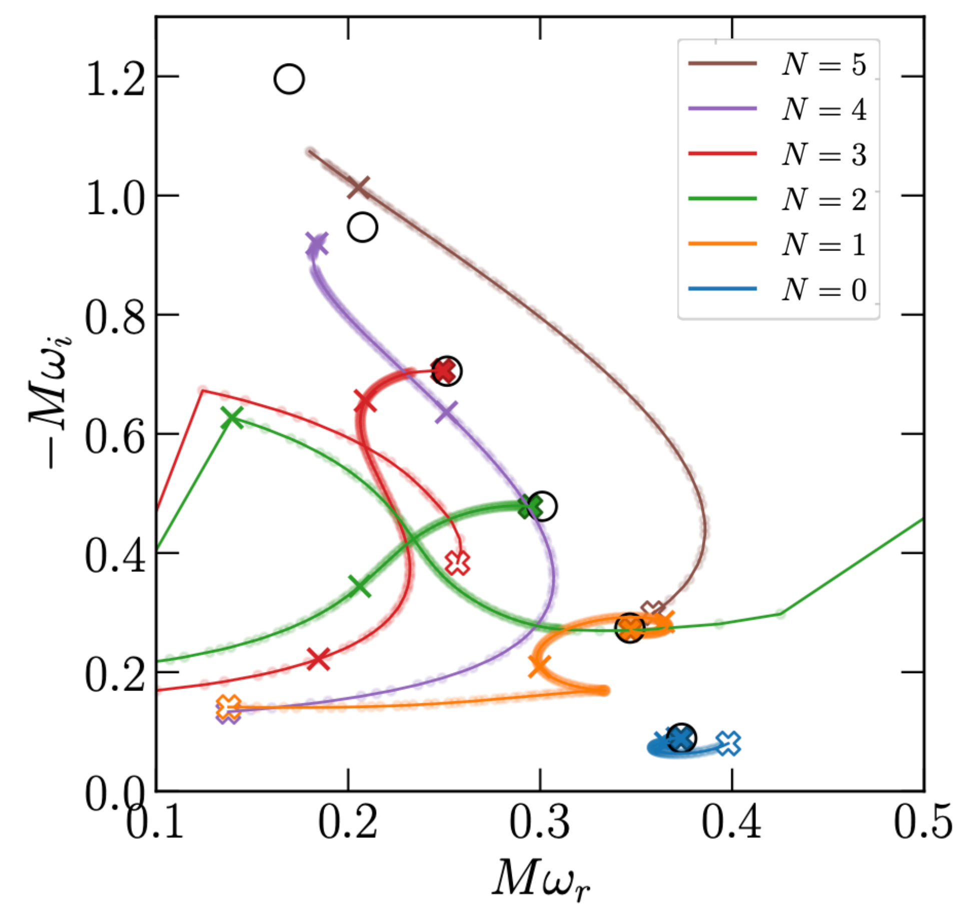
Extracting overtones in linearized theory



The authors use three toy waveforms: (1) a $\mathbf{Q_7}(t)$ composed entirely of damped sinusoids, (2) a waveform $\mathbf{Q_7^T}(t)$ with contamination from a power-law tail, and (3) a waveform $\boldsymbol{\Psi}_{num}(t)$ obtained numerically via linear perturbation theory.

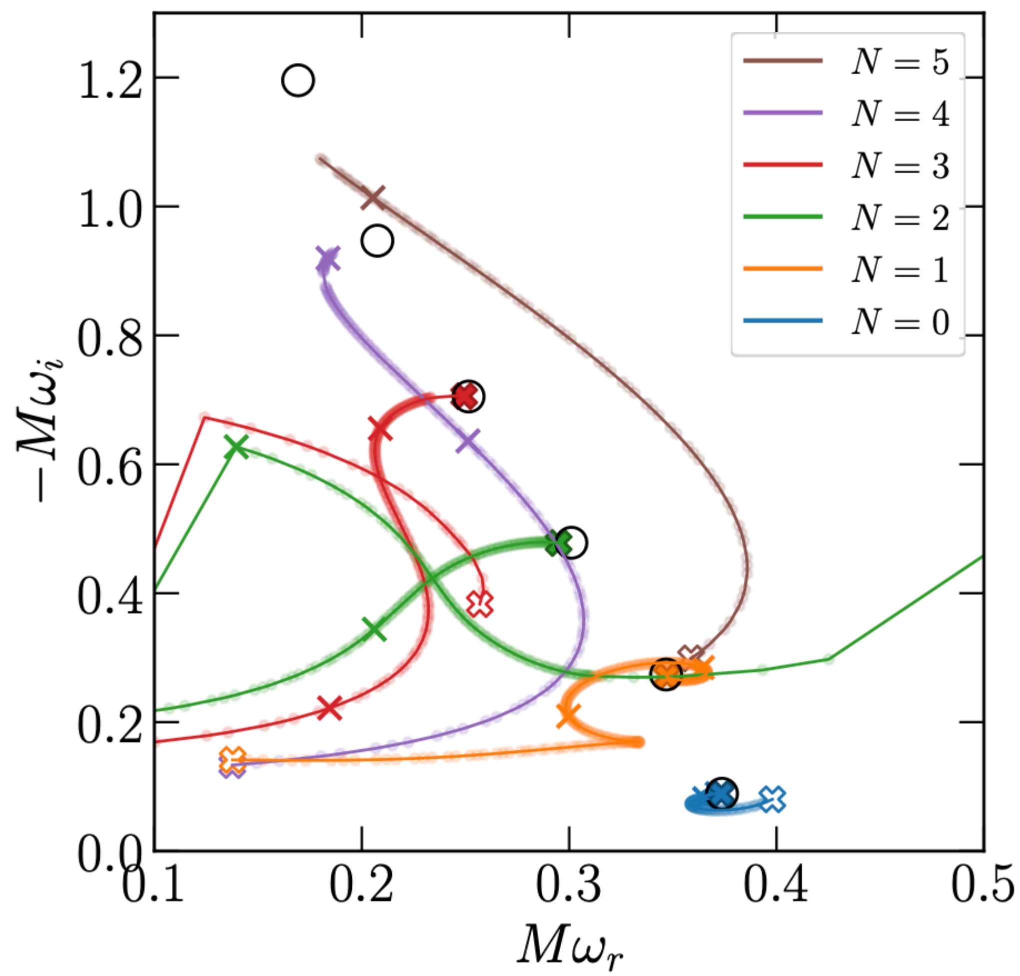
Extracting overtones in linearized theory

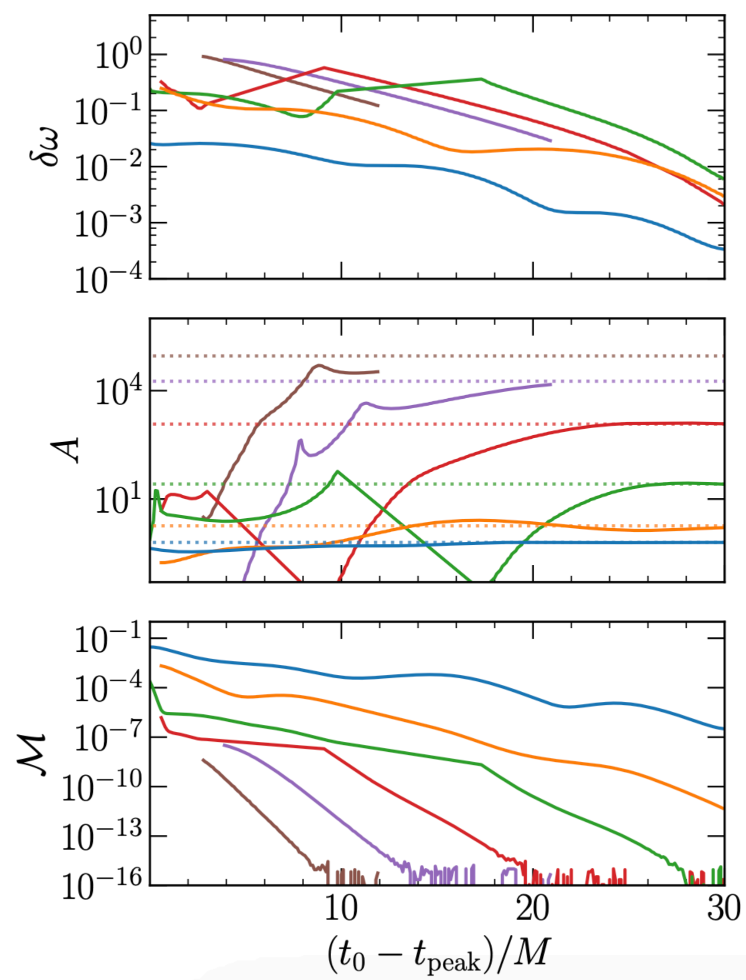




The extracted complex frequencies depend on the fitting window, with later-time fits generally doing a better job of converging to the actual value.

Extracting overtones in linearized theory





Deviation of free fitted frequency from expected overtone frequency

$$\delta\omega = M|\,\omega_r + i\omega_i - \omega_{ref}\,|$$

Amplitude of the free-frequency mode, obtained by extrapolating backward to t_{peak}. Actual amplitudes drawn as dotted lines

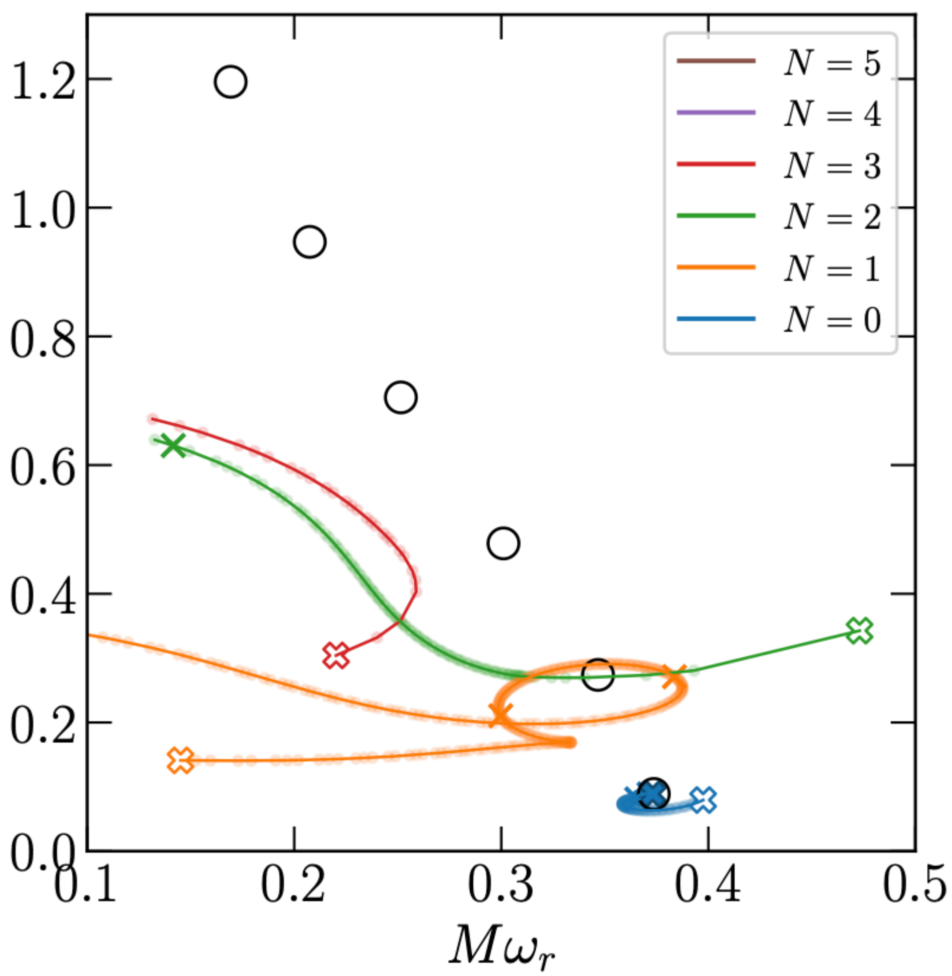
Mismatch between best-fit model and actual waveform

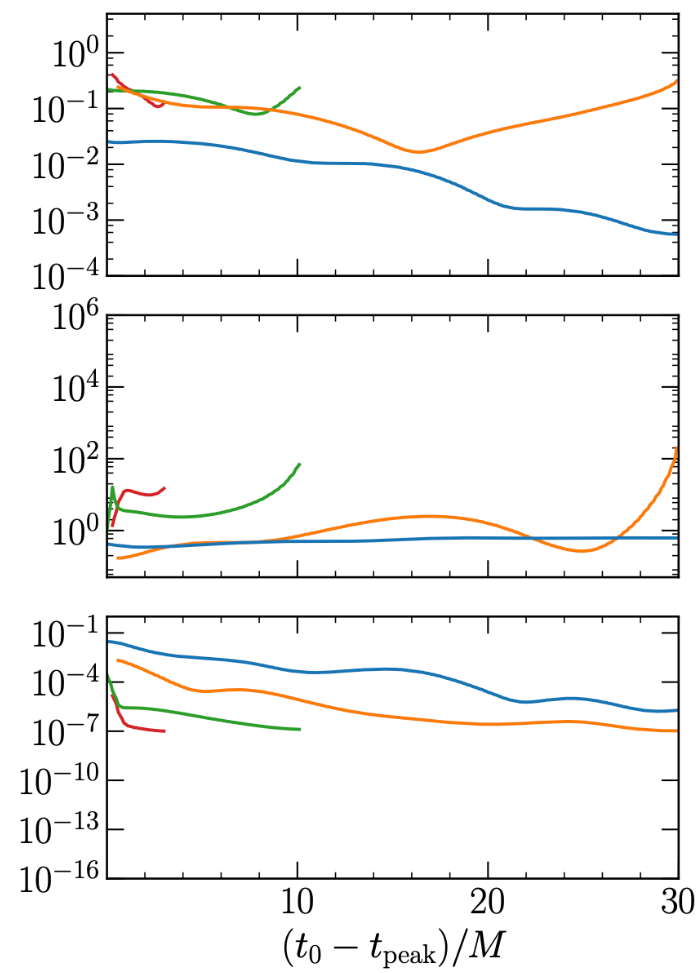
$$\mathscr{M} = 1 - \frac{\langle \Psi \,|\, \psi \rangle}{\sqrt{\langle \Psi \,|\, \Psi \rangle \langle \psi \,|\, \psi \rangle}},$$

$$\langle f \,|\, g \rangle = \int_{t_0}^{t_{end}} f(t)\overline{g(t)}\, dt\,.$$

The "pure ringdown" waveform $\mathbf{Q_7}(t)$ is well-modelled by the frequency-agnostic fits for $N \lesssim 3$ overtones.

Extracting overtones in linearized theory





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Mismatch between best-fit model and actual waveform

$$\mathscr{M} = 1 - \frac{\langle \Psi \,|\, \psi \rangle}{\sqrt{\langle \Psi \,|\, \Psi \rangle \langle \psi \,|\, \psi \rangle}},$$

$$\langle f \,|\, g \rangle = \int_{t_0}^{t_{end}} f(t)\overline{g(t)}\, dt\,.$$

The fit to the physical waveform $\boldsymbol{\Psi}_{num}(t)$ is much worse – *only the fundamental mode can be recovered by an agnostic fit.* Fits with $N > 3$ were omitted by the authors in this plot due to poor convergence.

Extracting overtones in linearized theory

SUMMARY

- While the purely agnostic fit does a good job of recovering a high number of overtones $(N \lesssim 3)$ for a pure superposition of QNMs, this gets worse as we add theoretically predicted contamination (a late time power-law tail).

- For the full numerical waveform $\boldsymbol{\Psi}_{num}(t)$, only the fundamental overtone can be recovered. Recovery of the first overtone depends sensitively on the fitting window used; in particular, it is not true that late-time fits do a better job of converging to the predicted frequency.

- In all cases, increasing the number of overtones decreases the mismatch $\mathscr{M}$. A small mismatch is a *necessary but not sufficient* condition to conclude that the fitting model is consistent with the actual waveform.

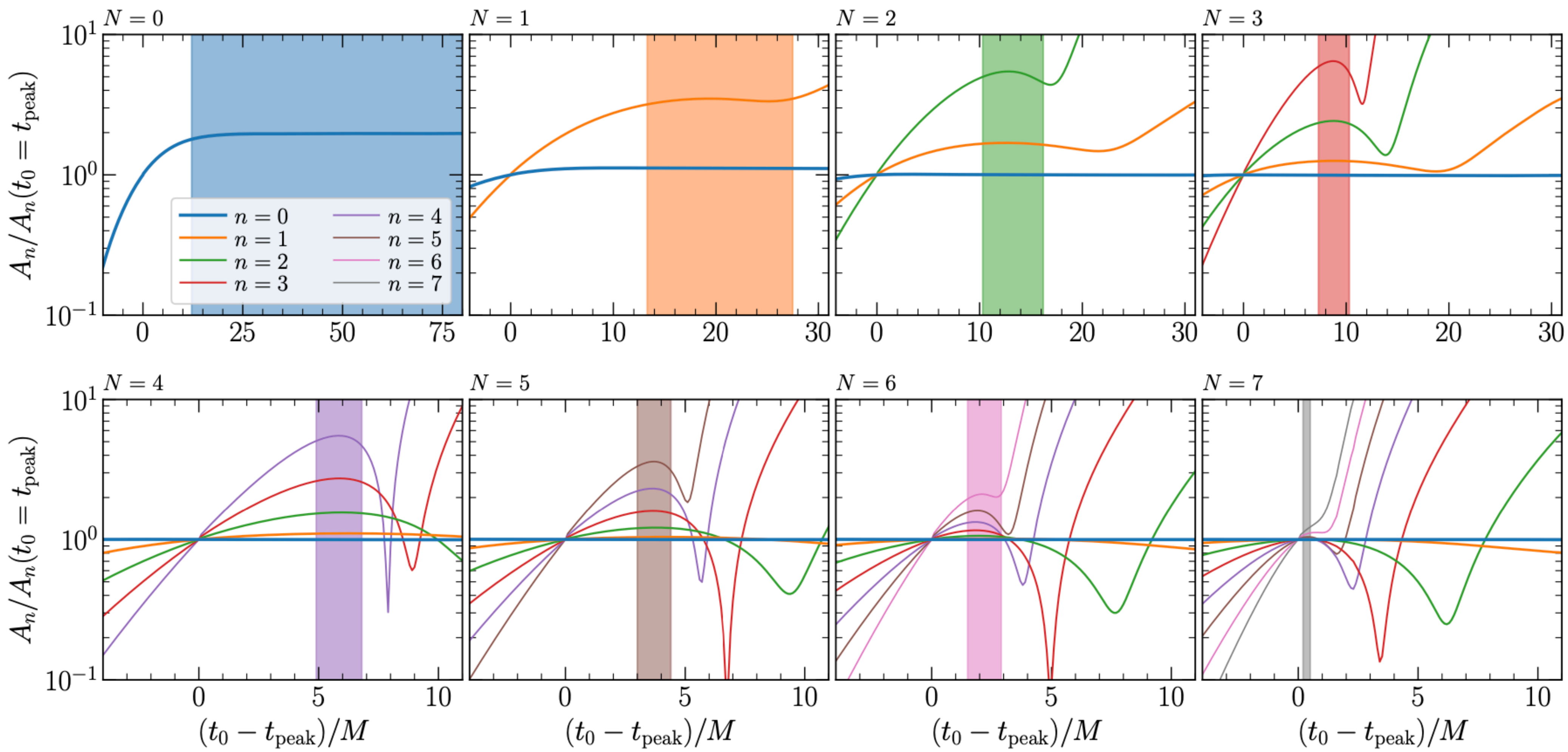
Are post-peak BBH waveforms linear?

- If linear perturbation theory is indeed a good description of the post-peak waveform, then the extracted QNM amplitudes at $t = t_{peak}$ should not depend on the fitting window used.

- If the higher overtones obtained via fitting are actually saying something about the physical content of the waveform, then generally,

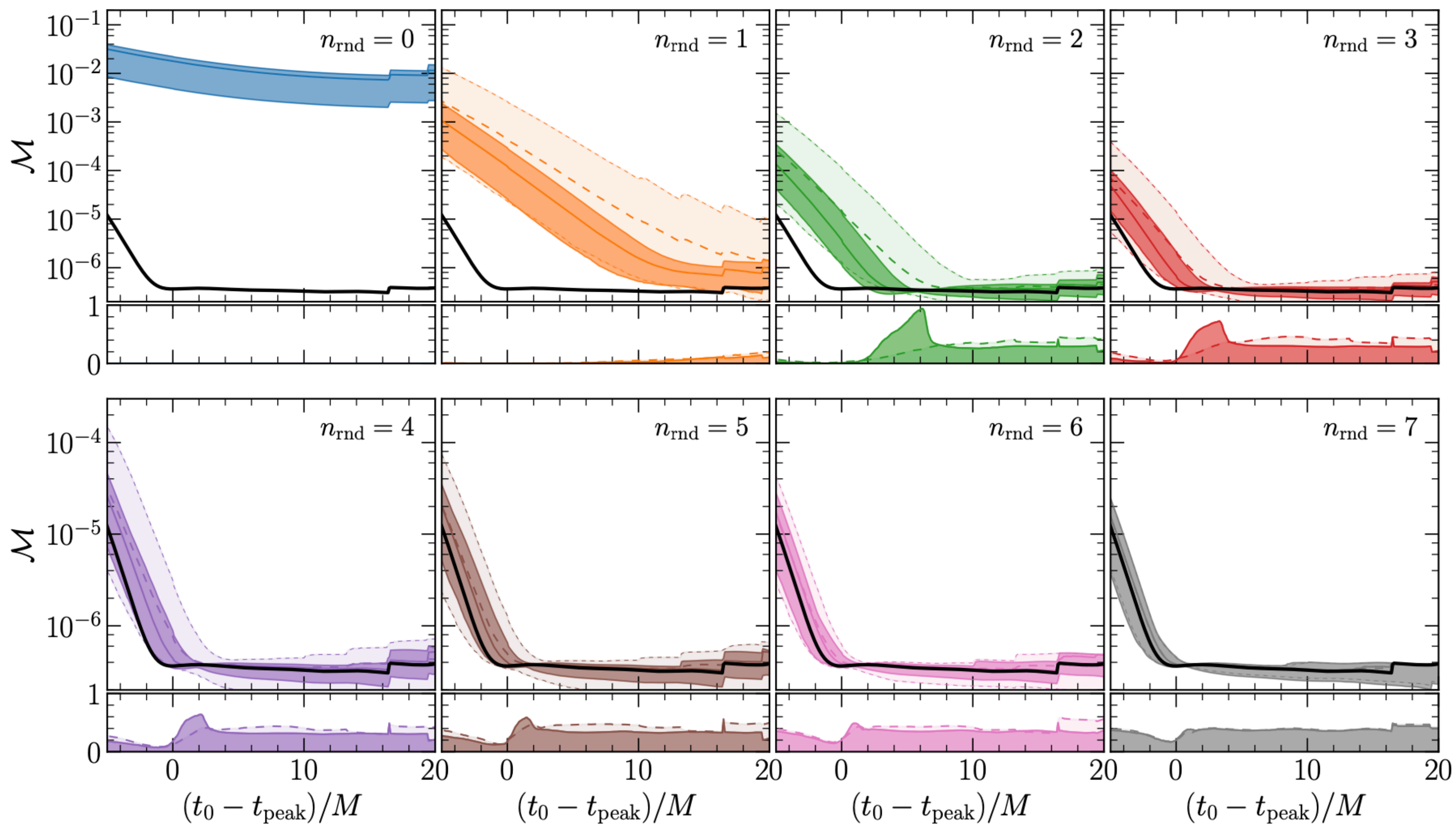
    - the mismatch should not improve if we replace several high-order overtones by modes with randomly chosen (i.e. unphysical) frequencies ω_n .
    - the parameter extraction should not improve due to the same replacement.

Are post-peak BBH waveforms linear?



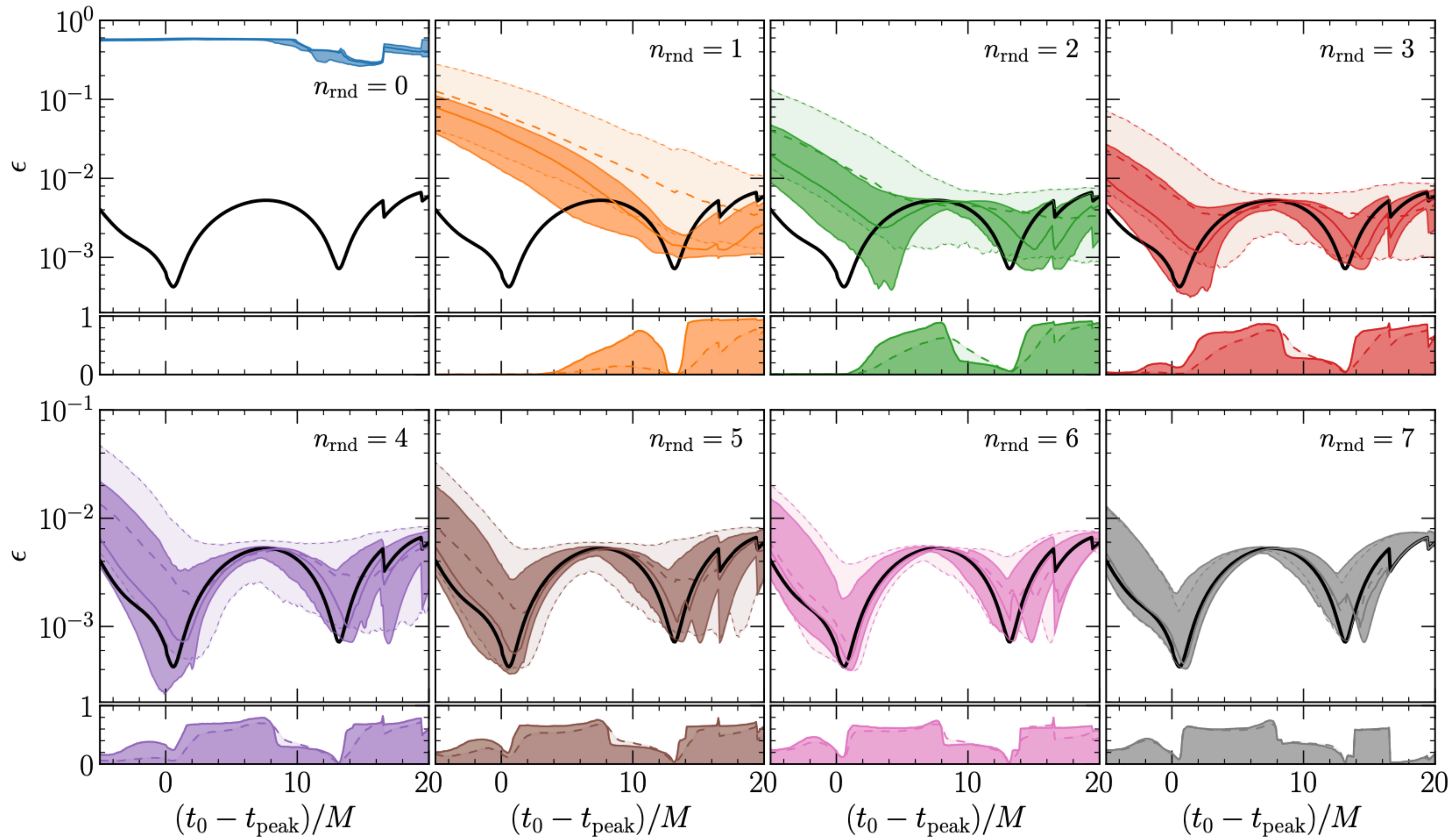
The colored regions show the range over which the amplitude of the highest overtone is constant to within 10%. Since the highest overtones never saturate to a constant value, this suggests that they do not actually contribute to the signal close to the peak, and merely overfit unmodelled contributions to the early signal.

Are post-peak BBH waveforms linear?



For higher overtones, a randomly chosen overtone frequency is about as good as the purported "physical" frequency at reproducing the mismatch with the NR waveform.

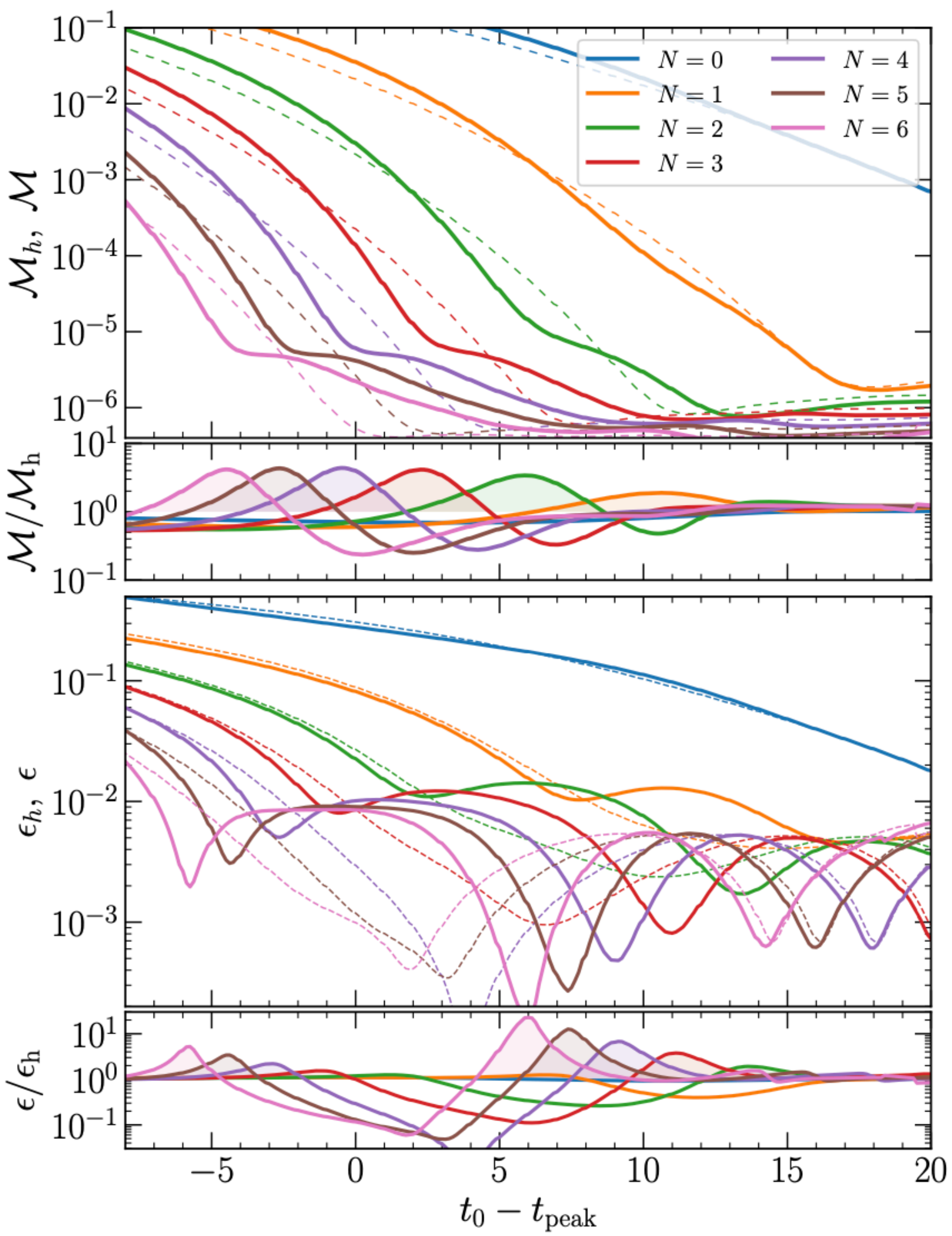
Are post-peak BBH waveforms linear?



$$\epsilon = \sqrt{(\delta\chi_f)^2 + \left(\frac{\delta M_f}{M}\right)^2}$$

Similarly, a random choice of higher overtones still gives a good estimate of the remnant parameters. A significant number of the random samples actually result in a better fit than the "physical" frequencies!

Are post-peak BBH waveforms linear?



In Appendix E, the authors look at two NR waveforms h_1 and h_2, smoothly blended together into an unphysical hybrid waveform that is SXS:BBH:0220 at $t = t_{peak}$, a combination of the two for intermediate times, and is SXS:BBH:0305 for $t \geq t_{peak} + 20M$.

The hybrid waveform is fitted using the frequencies of the second waveform; this has no theoretical basis. In spite of this, including more overtones still manages to reduce the mismatch and parameter extraction (from the 2nd waveform.)

Thus, the role of higher overtones is to fit the early post-peak waveform, which allows for better extraction of the fundamental mode. This explains the improvement in ϵ .

Are post-peak BBH waveforms linear?

CONCLUSION

Can we reliably conclude from an analysis of NR simulations that the entire post-peak waveform is described by a superposition of overtones consistent with linear BH perturbation theory of a fixed Kerr background?

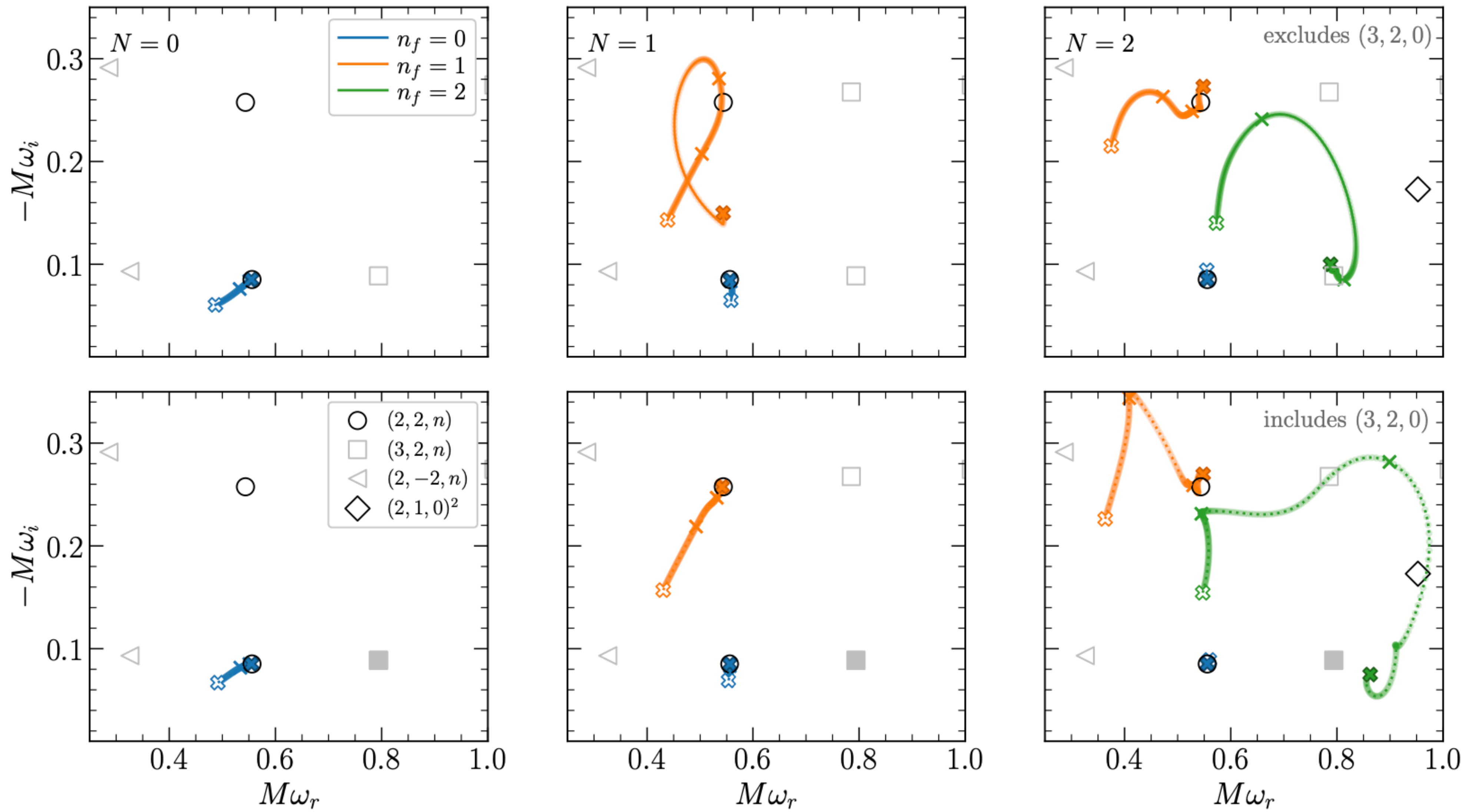
No, for the following reasons:

(1) The QNM amplitudes obtained from the fit with high overtones are inconsistent with the predictions of linearized theory.

(2) We can get comparably good mismatch and parameter extraction by using random overtone frequencies, instead of assuming $\omega_{\ell mn} = \omega_{\ell mn}(M_f, \chi_f)$.

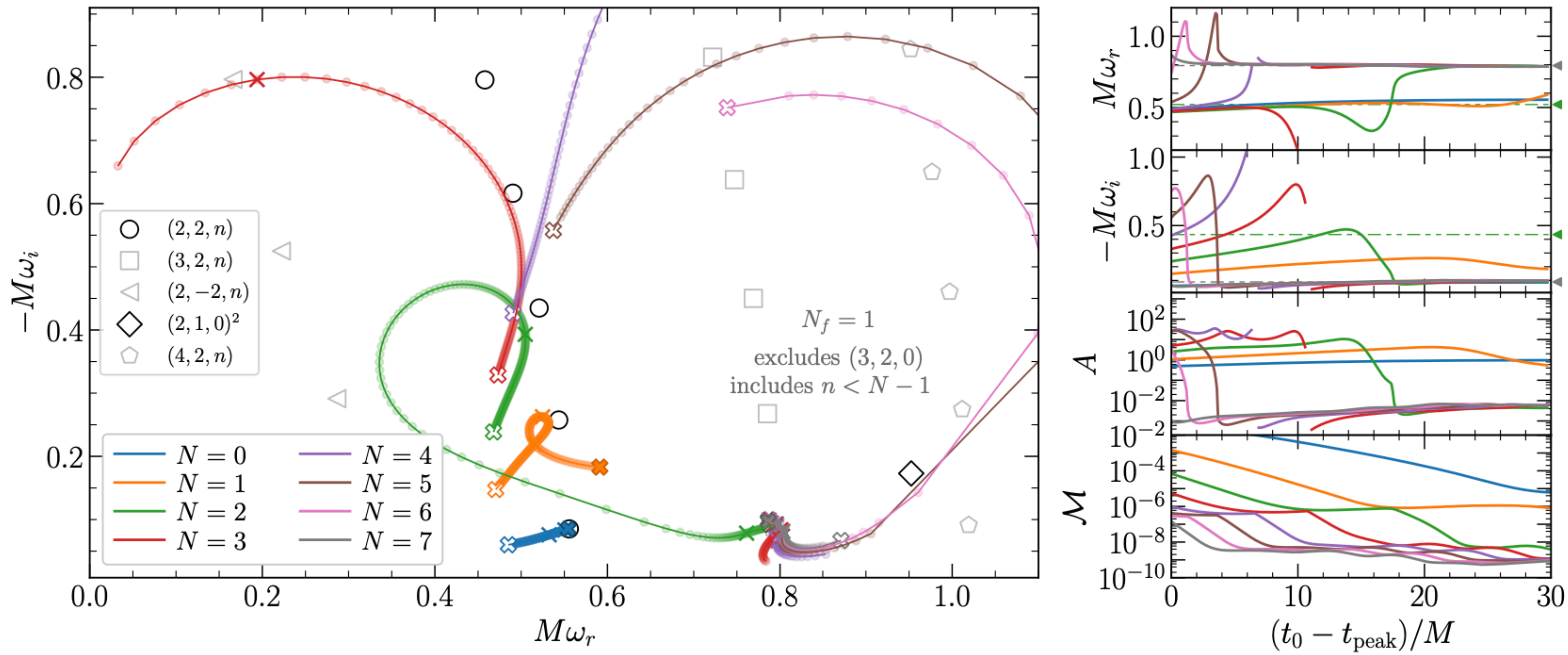
(3) The high-overtone fit still works when replacing the early post-peak waveform with "junk" data, suggesting higher overtones merely overfit early-time data.

Extracting complex frequencies from the waveform



Instead of assuming *a priori* that certain modes are present in the data (which can lead to overfitting), the authors perform an agnostic search. When assuming pure $(\ell, m) = (2,2)$, the $n = 2$ overtone latches on to the $(3,2,0)$ mode. "Subtracting" the contribution from mode-mixing by including the $(3,2,0)$ mode in the fit leads to better results for $N = 1$.

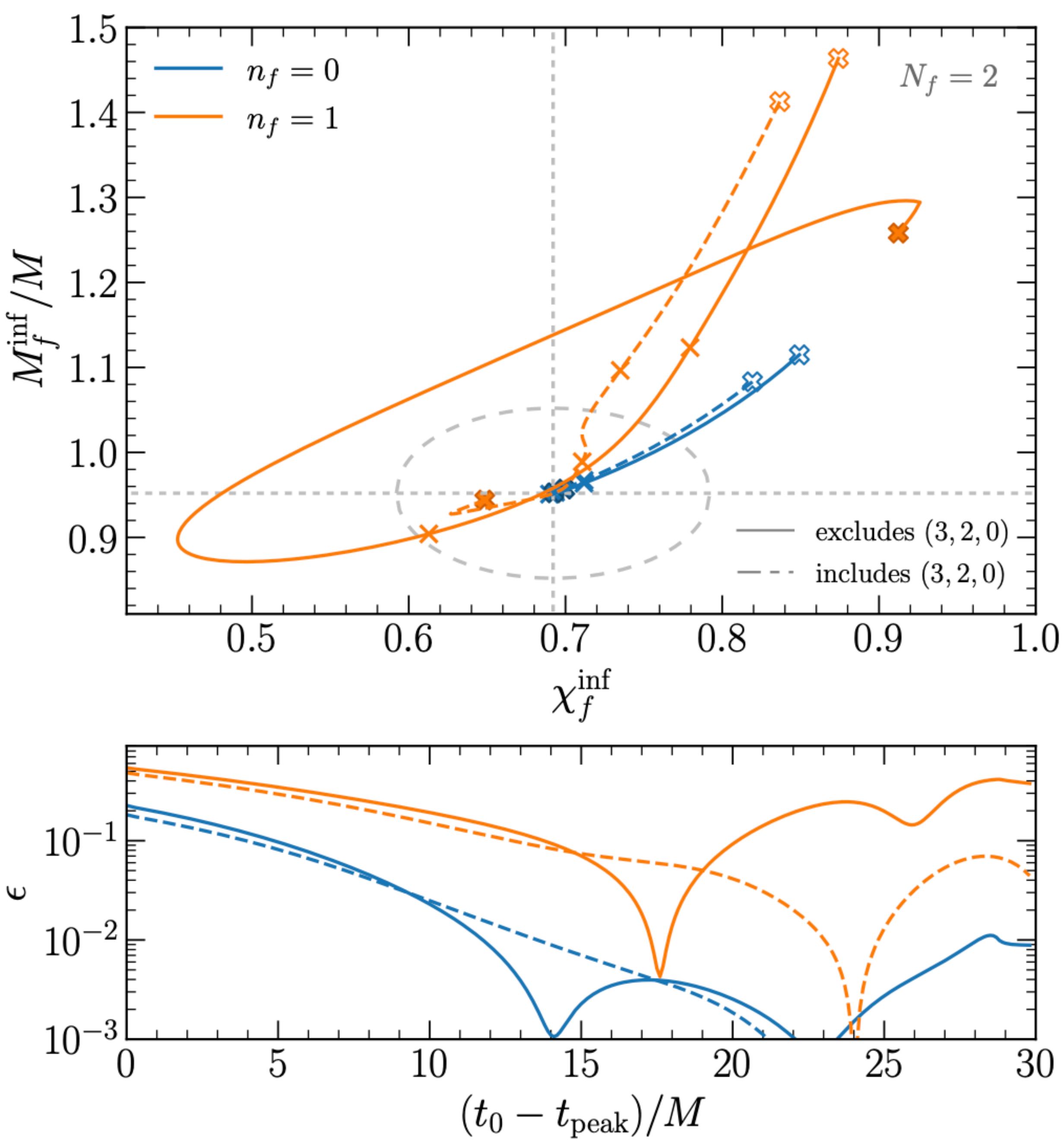
Extracting complex frequencies from the waveform



To show that this isn't an artifact of the free-frequency model, the authors examine $Q_{N,1}(t)$ fits, where N modes are fixed to their predicted values, and one mode is left free. As before, only the fundamental mode can be recovered. Interestingly, for the $N > 1$ models, the free modes consistently latch on to the $(3,2,0)$ mode.

Lessons: Generally, one cannot robustly identify high $(N > 1)$ overtones. Further, it is insufficient to consider only the $(\ell, m) = (2,2)$ mode; the $(3,2,0)$ mode also matters!

Extracting complex frequencies from the waveform



- Suppose we wish to infer the mass and spin of the remnant in the spirit of the BH spectroscopy program. Then we perform an *agnostic* fit and, *a posteriori,* assume that these modes correspond to $(2, 2, 0)$ and $(2, 2, 1)$.

- The addition of mode-mixing somewhat improves parameter recovery from the first overtone.

- Results are good for the fundamental mode, but not when the first overtone is included. *Even in the infinite SNR limit, it is difficult to get percent-level estimates of the remnant parameters with $n = 1$.*

Conclusions and discussion

- *The inclusion of a large number of overtones in BH spectroscopy has no physical justification*.

- High-overtone fits do not necessarily indicate physical excitation of higher modes of the system, but merely overfit the early-time signal, which in turn allows for better extraction of the fundamental mode.

- Overtone recovery may be poor due to spectral instability; there is a sense in which the fundamental mode is stabler under certain classes of perturbation to the potential.

Thank you!